

1.

A	B	C	$A'B$	$B'C'$	AC	F
0	0	0	0	1	0	1
0	0	1	0	0	0	0
0	1	0	1	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	1
1	0	1	0	0	1	1
1	1	0	0	0	0	0
1	1	1	0	0	1	1

2.

$$m_0 = A'B'C', \quad m_2 = A'BC', \quad m_5 = AB'C, \quad m_7 = ABC$$

$$\boxed{F = A'B'C' + A'BC' + AB'C + ABC}$$

3.

$$M_1 = (A + B + C'), \quad M_3 = (A + B' + C'),$$

$$M_4 = (A' + B + C), \quad M_6 = (A' + B' + C)$$

$$\boxed{F = (A + B + C')(A + B' + C')(A' + B + C)(A' + B' + C)}$$

4.

$$F' = [x'y + z(w + x)]'$$

$$= (x'y)' [z(w + x)]'$$

$$= (x + y')(z' + (w + x)')$$

$$\boxed{F' = (x + y')(z' + w'x')}$$

5.

$$F = A'BC + A'BC' + AB'C' + ABC'$$

$$= A'B(C + C') + AC'(B' + B)$$

$$= A'B \cdot 1 + AC' \cdot 1$$

$$\boxed{F = A'B + AC'}$$

6.

$$G = (x + z)(x + z')(x' + y)$$

$$= (x + zz')(x' + y)$$

$$= x(x' + y)$$

$$= xx' + xy = 0 + xy$$

$$\boxed{G = xy}$$

7.

$$H = w'x'y + w'xy + wx'y + wxy$$

$$= y(w'x' + w'x + wx' + wx)$$

$$= y[w'(x' + x) + w(x' + x)]$$

$$= y(w' \cdot 1 + w \cdot 1) = y(w' + w)$$

$$\boxed{H = y}$$

8.

$$\begin{aligned} & pq + p'r + qr \\ &= pq + p'r + qr(p + p') \\ &= pq + p'r + pqr + p'qr \\ &= pq(1 + r) + p'r(1 + q) \\ &= pq \cdot 1 + p'r \cdot 1 \end{aligned}$$

$$\boxed{pq + p'r + qr = pq + p'r \text{ is true.}}$$

9.

$$\begin{aligned} P &= 01011100 \\ Q &= 11001010 \\ P + Q &= 11011110 \\ (P + Q)' &= 00100001 \end{aligned}$$

$$\boxed{P \downarrow Q = 00100001}$$

10.

$$Y = (A' + B)(C + D') + AC$$